1. **Check Even or Odd:** Write a program to determine whether a given number is Even or Odd.

```cpp
// 01_even_odd.cpp
// Check if a number is Even or Odd
#include <iostream>
using namespace std;
int main() {
int num;
cout << "Enter a number: ";
cin >> num;
if (num % 2 == 0)
cout << num << " is Even." << endl;
else
cout << num << " is Odd." << endl;
return 0;
}


```

```
mac@SimmiTiwari c++_college % g++ Assignment2.cpp -o Assignment2
./Assignment2
Enter a number: 7
7 is Odd.
mac@SimmiTiwari c++_college % ./Assignment2
Enter a number: 67
67 is Odd.
mac@SimmiTiwari c++_college %
```

2. **Largest Among Three Numbers**: Write a program to find the largest among three given nos.

ASSIGNMENT2 > Q2.cpp > main()

```cpp
// 02_largest_of_3.cpp
// Find the Largest Among 3 Numbers
#include <iostream>
using namespace std;
int main() {
int a, b, c;
cout << "Enter three numbers: ";
cin >> a >> b >> c;
if (a >= b && a >= c)
cout << "Largest number is: " << a << endl;
else if (b >= a && b >= c)
cout << "Largest number is: " << b << endl;

else
cout << "Largest number is: " << c << endl;

return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q2
Enter three numbers: 23 56 66
Largest number is: 66
mac@SimmiTiwari ASSIGNMENT2 % ./Q2
Enter three numbers: 67 88 1000
Largest number is: 1000
mac@SimmiTiwari ASSIGNMENT2 %
```

3. **Vowel or Consonant Check:** Write a program to check whether the entered alphabet is a vowel or a consonant.

ASSIGNMENT2 > Q3.cpp > main()

```cpp
// 03_vowel_consonant.cpp
// Check if a character is Vowel or Consonant
#include <iostream>
using namespace std;
int main() {
char ch;
cout << "Enter a character: ";
cin >> ch;
// Convert to lowercase for uniform comparison
ch = tolower(ch);
if (ch >= 'a' && ch <= 'z') {
if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
cout << ch << " is a Vowel." << endl;
else
cout << ch << " is a Consonant." << endl;
} else {
cout << ch << " is not an alphabet character." << endl;
}
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % ./Q3
Enter a character: Dhruv
d is a Consonant.
mac@SimmiTiwari ASSIGNMENT2 %
```

4. **Leap Year Check**: Write a program to determine whether a given year is a Leap Year.

Q1.cpp Q2.cpp Q3.cpp Q4.cpp

ASSIGNMENT2 > Q4.cpp > main()

```cpp
// 04_leap_year.cpp
// Check if a Year is a Leap Year
#include <iostream>
using namespace std;
int main() {
int year;
cout << "Enter a year: ";
cin >> year;
if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
cout << year << " is a Leap Year." << endl;
else
cout << year << " is NOT a Leap Year." << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % ./Q4
Enter a year: 1994
1994 is NOT a Leap Year.
mac@SimmiTiwari ASSIGNMENT2 %
```

5. **Multiplication Table:** Write a program to display the multiplication table of a given number form

```
ASSIGNMENT2 > Q5.cpp > main()
// 05_multiplication_table.cpp
// Print the Multiplication Table of a Number
#include <iostream>
using namespace std;

int main() {
int num;
cout << "Enter a number: ";
cin >> num;
cout << "\nMultiplication Table of " << num << ":" << endl;
cout << "------------------------------" << endl;
for (int i = 1; i <= 10; i++) {
cout << num << " x " << i << " = " << num * i << endl;
}
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q5
Multiplication Table of 34:
-----------------------------
34 x 1 = 34
34 x 2 = 68
34 x 3 = 102
34 x 4 = 136
34 x 5 = 170
34 x 6 = 204
34 x 7 = 238
34 x 8 = 272
34 x 9 = 306
34 x 10 = 340
mac@SimmiTiwari ASSIGNMENT2 %
```

6. **Sum of First N Natural Numbers:** Write a program to calculate the sum of the first .

```
ASSIGNMENT2 > Q6.cpp > main()
// 06_sum_of_n_naturals.cpp
// Sum of First N Natural Numbers
#include <iostream>
using namespace std;
int main() {
int n;
cout << "Enter the value of N: ";
cin >> n;
if (n < 0) {
cout << "Please enter a positive integer." << endl;
return 1;
}
long long sum = 0;
for (int i = 1; i <= n; i++) {
sum += i;
}
cout << "Sum of first " << n << " natural numbers = " << sum << endl;
cout << "(Using formula: " << n << " * " << (n + 1) << " / 2 = " << (long long)n * (n + 1) / 2 << ")" <<
endl;

return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q6
Enter the value of N: 5
Sum of first 5 natural numbers = 15
(Using formula: 5 * 6 / 2 = 15)
```

7. **Factorial of a Number:** Write a program to calculate the factorial of a given number.

ASSIGNMENT2 > Q7.cpp > main()

```cpp
// 07_factorial.cpp
// Factorial of a Number
#include <iostream>
using namespace std;
int main() {
int n;
cout << "Enter a non-negative integer: ";
cin >> n;
if (n < 0) {
cout << "Factorial is not defined for negative numbers." << endl;
return 1;
}
long long factorial = 1;
for (int i = 1; i <= n; i++) {
factorial *= i;
}
cout << "Factorial of " << n << " (" << n << "!) = " << factorial << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q7
Enter a non-negative integer: 8
Factorial of 8 (8!) = 40320
```

8. **Reverse a Number**: Write a program to reverse the digits of a given number.

ASSIGNMENT2 > Q8.cpp > main()

```cpp
// 08_reverse_number.cpp
// Reverse a Number
#include <iostream>
using namespace std;
int main() {
long long num;
cout << "Enter a number: ";
cin >> num;
long long original = num;
long long reversed = 0;
bool isNegative = false;
if (num < 0) {
isNegative = true;
num = -num;
}
while (num != 0) {
int digit = num % 10;
reversed = reversed * 10 + digit;
num /= 10;
}
if (isNegative)
reversed = -reversed;
cout << "Original number : " << original << endl;
cout << "Reversed number : " << reversed << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q8
Enter a number: 42
Original number : 42
Reversed number : 24
```

9. **Greatest Common Divisor (GCD):** Write a program to find the Greatest Common Divisor (GCD) of two numbers

ASSIGNMENT2 > Q9.cpp > main()

```
// 09_gcd.cpp
// GCD (Greatest Common Divisor) of Two Numbers
// Uses the Euclidean Algorithm
#include <iostream>
using namespace std;
int gcd(int a, int b) {
while (b != 0) {
int temp = b;
b = a % b;
a = temp;
}
return a;
}
int main() {
int a, b;
cout << "Enter two positive integers: ";
cin >> a >> b;
if (a <= 0 || b <= 0) {
cout << "Please enter positive integers." << endl;
return 1;
}
cout << "GCD of " << a << " and " << b << " = " << gcd(a, b) << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q9
Enter two positive integers: 12 15
GCD of 12 and 15 = 3
```

10. **Least Common Multiple (LCM)**: Write a program to find the Least Common Multiple (LCM) of two numbers.

ASSIGNMENT2 > Q10.cpp > main()

```
// 10_lcm.cpp
// LCM (Least Common Multiple) of Two Numbers
// Uses the relation: LCM(a, b) = (a * b) / GCD(a, b)
#include <iostream>
using namespace std;
int gcd(int a, int b) {
while (b != 0) {
int temp = b;
b = a % b;
a = temp;
}
return a;
}
long long lcm(int a, int b) {
return ((long long)a / gcd(a, b)) * b;
}
int main() {
int a, b;
cout << "Enter two positive integers: ";
cin >> a >> b;
if (a <= 0 || b <= 0) {
cout << "Please enter positive integers." << endl;
return 1;
}
cout << "GCD of " << a << " and " << b << " = " << gcd(a, b) << endl;
cout << "LCM of " << a << " and " << b << " = " << lcm(a, b) << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q10
Enter two positive integers: 15 18
GCD of 15 and 18 = 3
LCM of 15 and 18 = 90
```

11. **Palindrome Number Check**: Write a program to determine whether a given number is a palindrome.

ASSIGNMENT2 > Q11.cpp > main()

```cpp
// 11_palindrome.cpp
// Check if a Number is a Palindrome
#include <iostream>
using namespace std;
int main() {
long long num;
cout << "Enter a number: ";
cin >> num;
if (num < 0) {
cout << num << " is NOT a Palindrome (negative numbers are not palindromes)." << endl;
return 0;
}
long long original = num;
long long reversed = 0;
long long temp = num;
while (temp != 0) {
int digit = temp % 10;
reversed = reversed * 10 + digit;
temp /= 10;
}
if (original == reversed)
cout << original << " IS a Palindrome." << endl;
else
cout << original << " is NOT a Palindrome." << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q11
Enter a number: 196
196 is NOT a Palindrome.
```

12. **Prime Number Check:** Write a program to determine whether a given number is a Prime Number.

ASSIGNMENT2 > Q12.cpp > main()

```cpp
// 12_prime_check.cpp
// Check if a Number is Prime
#include <iostream>
#include <cmath>
using namespace std;
bool isPrime(int n) {
if (n <= 1) return false;
if (n <= 3) return true;
if (n % 2 == 0 || n % 3 == 0)
return false ;
for (int i = 5; i <= sqrt(n); i += 6) {
if (n % i == 0 || n % (i + 2) == 0)
return false;
}
return true;
}
int main() {
int num;
cout << "Enter a number: ";
cin >> num;
if (isPrime(num))
cout << num << " IS a Prime number." << endl;
else
cout << num << " is NOT a Prime number." << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q12
Enter a number: 12
12 is NOT a Prime number.
```

13. **Prime Numbers in a Range**: Write a program to display all Prime Numbers between two given numbers.

ASSIGNMENT2 > Q13.cpp > main()

```cpp
// 13_primes_in_range.cpp
// Print all Prime Numbers in a Given Range
#include <iostream>
#include <cmath>
using namespace std;
bool isPrime(int n) {
if (n <= 1) return false;
if (n <= 3) return true;
if (n % 2 == 0 || n % 3 == 0) return false;
for (int i = 5; i <= sqrt(n); i += 6) {
if (n % i == 0 || n % (i + 2) == 0)
return false;
}
return true;
}
int main() {
int low, high;
cout << "Enter the lower bound: ";
cin >> low;
cout << "Enter the upper bound: ";
cin >> high;
if (low > high) swap(low, high);
cout << "\nPrime numbers between " << low << " and " << high << ":" << endl;
int count = 0;
for (int i = low; i <= high; i++) {
if (isPrime(i)) {
cout << i << " ";
count++;
}
}
if (count == 0)
cout << "None found.";
cout << "\n\nTotal primes found: " << count << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q13
Enter the lower bound: 5
Enter the upper bound: 27

Prime numbers between 5 and 27:
5 7 11 13 17 19 23

Total primes found: 7
```

14. **Neon Numbers in a Range:** Write a program to display all Neon Numbers within a given range.

ASSIGNMENT2 > Q14.cpp > main()

```cpp
// 14_neon_numbers.cpp
// Find all Neon Numbers in a Given Range
#include <iostream>
using namespace std;
int sumOfDigits(long long n) {
int sum = 0;
while (n != 0) {
sum += n % 10;
n /= 10;
}
return sum;
}
bool isNeon(int n) {
long long square = (long long)n * n;
return sumOfDigits(square) == n;
}

int main() {
int low, high;
cout << "Enter the lower bound: ";
cin >> low;

cout << "Enter the upper bound: ";
cin >> high;
if (low > high) swap(low, high);
cout << "\nNeon numbers between " << low << " and " << high << ":" << endl;
int count = 0;
for (int i = low; i <= high; i++) {
if (isNeon(i)) {
long long sq = (long long)i * i;
cout << i << " (square = " << sq
<< ", sum of digits = " << sumOfDigits(sq) << ")" << endl;
count++;
}
}
if (count == 0)
cout << "No Neon numbers found in this range." << endl;
else
cout << "\nTotal Neon numbers found: " << count << endl;
return 0;
}
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q14
Enter the lower bound: 5
Enter the upper bound: 16

Neon numbers between 5 and 16:
9 (square = 81, sum of digits = 9)

Total Neon numbers found: 1
```

15. **Armstrong Number Check:** Write a program to determine whether a given number is an Armstrong Number.

ASSIGNMENT2 > Q15.cpp > main()

```cpp
// 15_armstrong_number.cpp
// Check if a Number is an Armstrong Number
#include <iostream>
#include <cmath>
using namespace std;

int countDigits(int n) {
int count = 0;
while (n != 0) {
count++;
n /= 10;
}
return count;
}
bool isArmstrong(int n) {
if (n < 0) return false;
int digits = countDigits(n);
int temp = n;
long long sum = 0;
while (temp != 0) {
int digit = temp % 10;
sum += (long long)pow(digit, digits);
temp /= 10;
}
return sum == n;
}
int main() {
int num;
cout << "Enter a number: ";
cin >> num;
if (num < 0) {
cout << "Armstrong numbers are non-negative." << endl;
return 1;
}
```

```cpp
    int digits = countDigits(num);
    int temp = num;
    long long sum = 0;

    cout << "\nNumber of digits: " << digits << endl;
    cout << "Calculation: ";
    bool first = true;
    while (temp != 0) {
    int digit = temp % 10;
    if (!first) cout << " + ";
    cout << digit << "^" << digits;
    sum += (long long)pow(digit, digits);
    temp /= 10;
    first = false;
    }
    cout << " = " << sum << endl;
    if (isArmstrong(num))
    cout << num << " IS an Armstrong number." << endl;
    else
    cout << num << " is NOT an Armstrong number." << endl;
    return 0;
    }
```

```
mac@SimmiTiwari ASSIGNMENT2 % /Users/mac/c++_college/ASSIGNMENT2/Q15
Enter a number: 34

Number of digits: 2
Calculation: 4^2 + 3^2 = 25
34 is NOT an Armstrong number.
```